The parts of the carriage are

a. Saddle
b. Cross slide
c. Compound rest
d. Compound slide
e. Tool post
f. Apron

Saddle

Saddle connects the front and back portion of bed guideways like a bridge. It is an 'H' shaped casting. It moves along the bed between headstock and tailstock. The saddle can be moved by providing hand feed or automatic feed.

Cross slide

Cross slide is situated on the saddle and slides on the dovetail guideways at right angles to the bed guideways. It carries compound rest, compound slide and tool post. The cross slide can be moved at right angles to the lathe axis by giving hand feed or automatic feed.

Compound rest

Compound rest is a part which connects cross slide and compound slide. It is mounted on the top of the cross slide by tongue and groove joint. It has a circular base on which angular graduations are marked. The compound rest can be swiveled to the required angle while turning tapers. A top slide known as compound slide is attached to the compound rest by dove tail joint. The tool post is fixed on the compound slide.

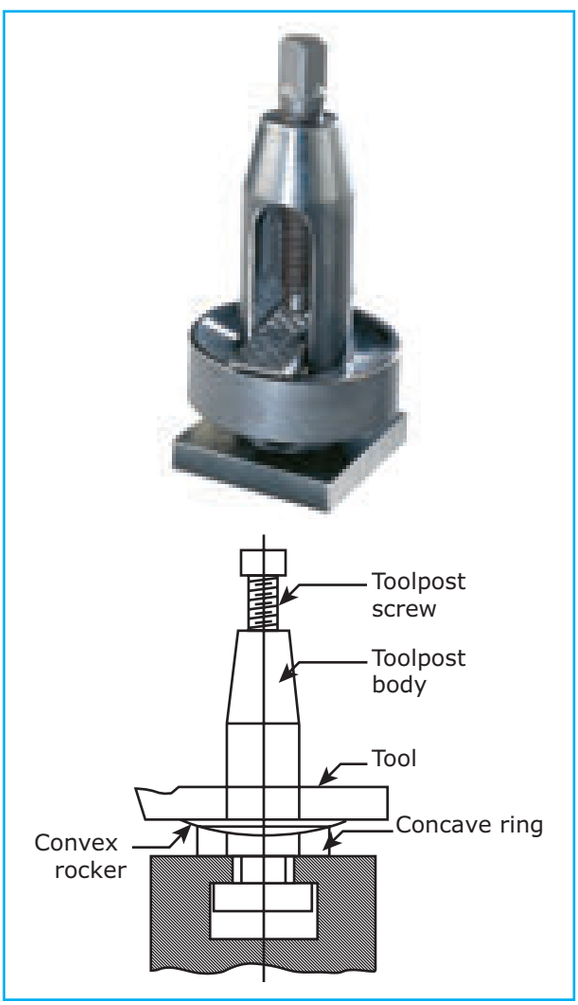
Tool post

The tool post is mounted on the compound slide. It is used for holding the cutting tool tightly. According to the operations on the work piece, tool may be changed and fixed on the tool post. There are different types of tool post. In which,

1. Single screw tool post
2. Four way tool post are widely used.

Single screw tool post

One tool can be held by the single screw tool post. The tool is clamped by one clamping screw. A concave ring is found in the tool post. A convex rocker iron piece has got a convex surface at its bottom and flat surface at its top. This rocker iron piece is placed on the concave ring. The tool rests on the top flat surface of the rocker.



Single screw tool post

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